NAU
NORTHERN ARIZONA UNIVERSITY

# Project RAPTOR

## Rocket Apogee Payload Transition to Optimal Return

## Midterm Progress Report

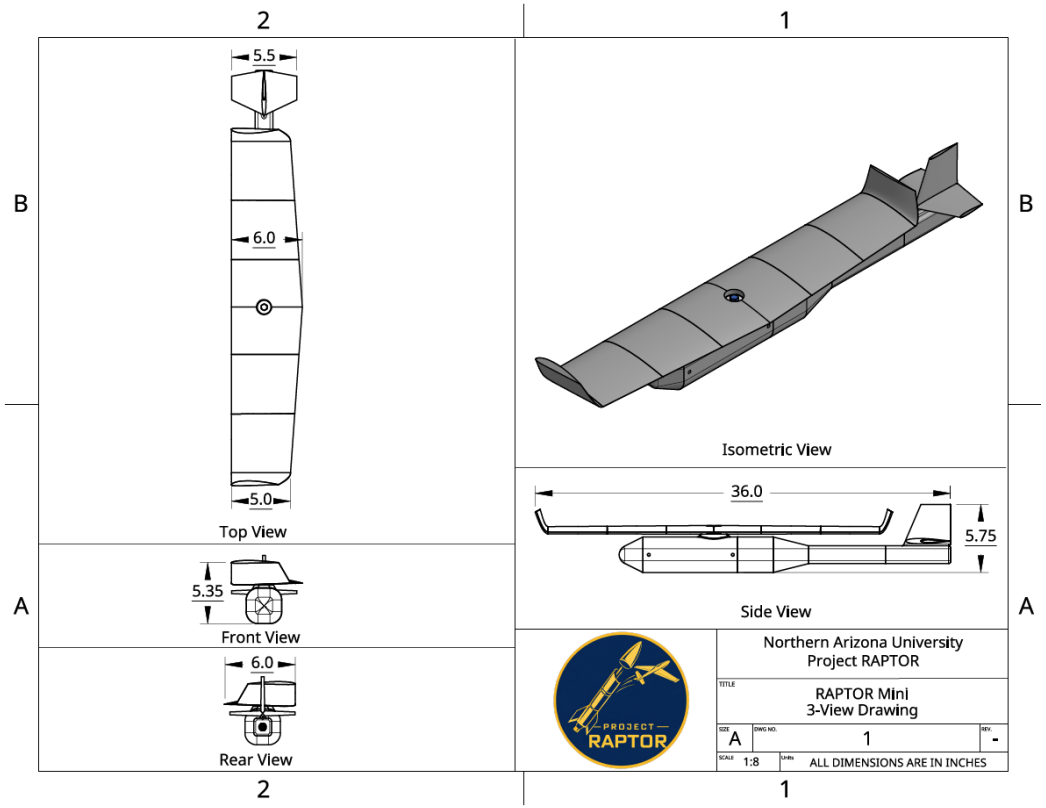
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**Date:** October 23, 2026

# RAPTOR Mini Progress

## Launch Stage Orientation

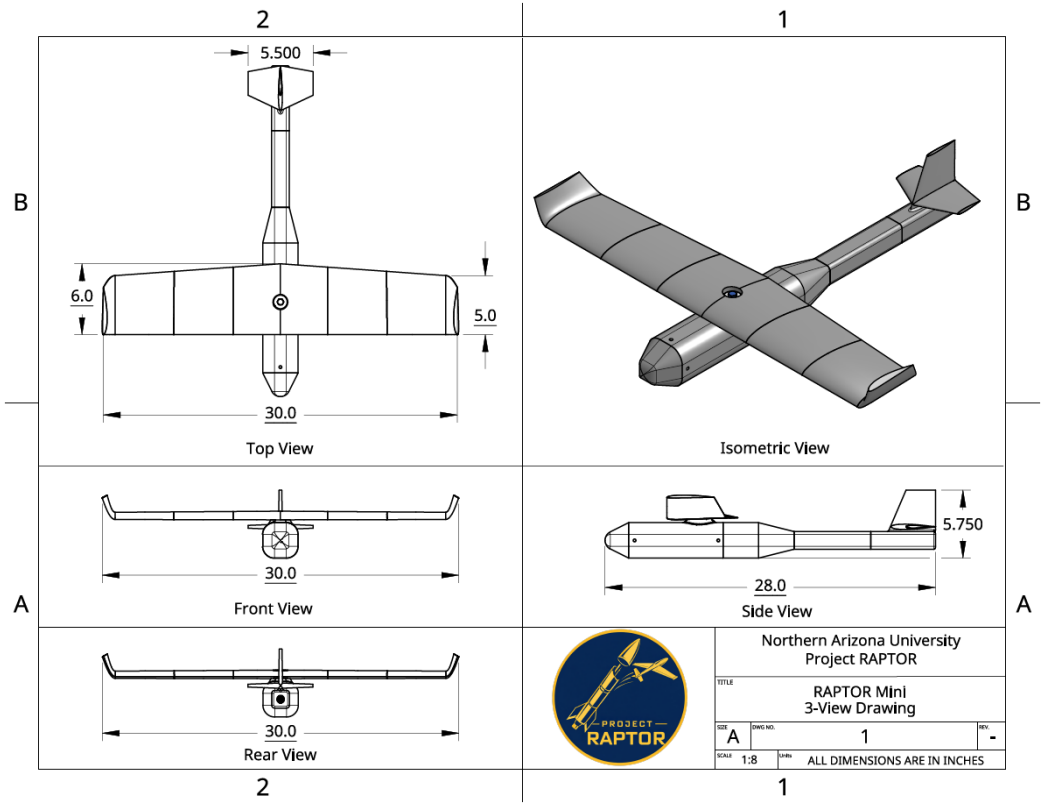
- Developed a 3D-printable CAD model from a customer requirements and engineering constraints
- Estimate maximum vehicle weight and sized wings to provide required lift
- Designed the airframe to package the wings in a folded configuration




[1] Figure 1: UAV 3-View Launch Orientation Drawing

## Flight Stage Orientation

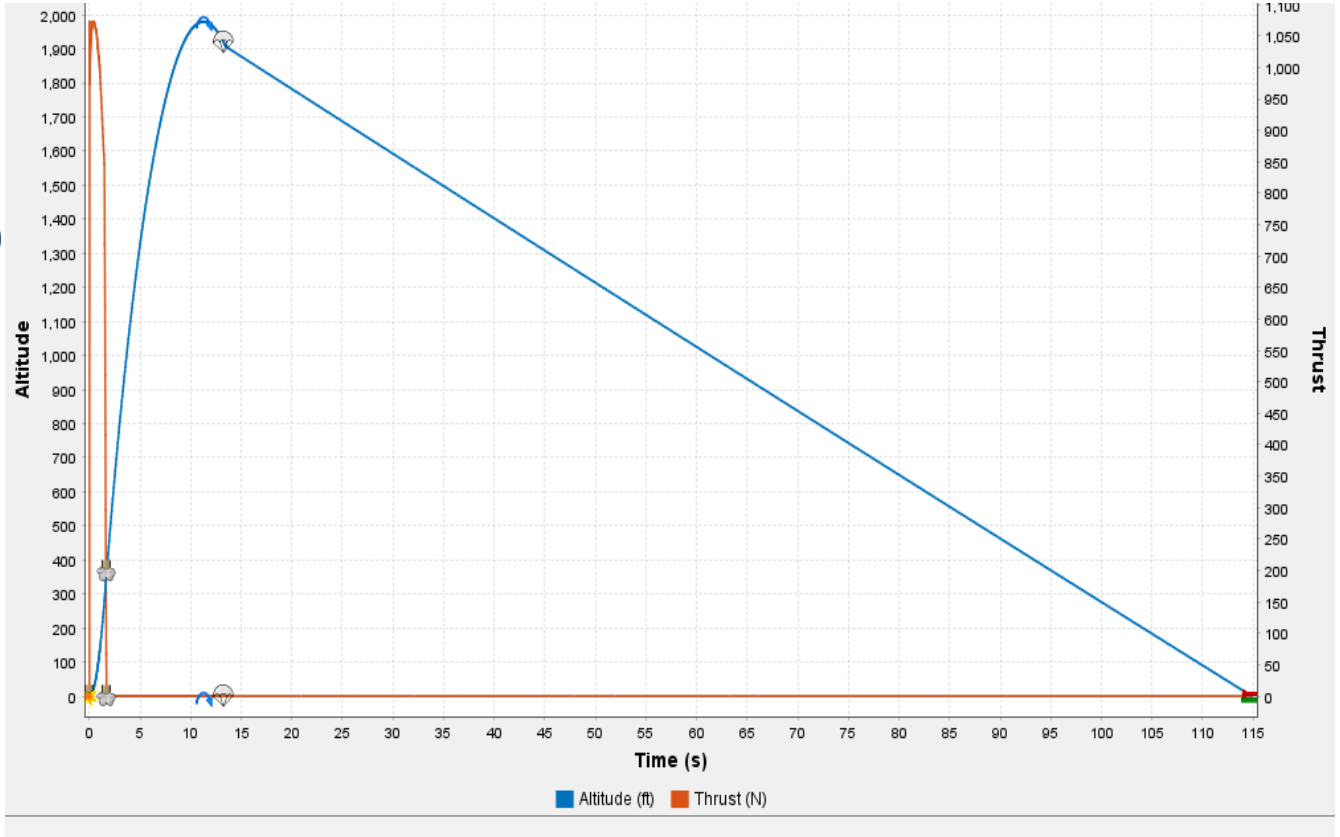
- Designed a folding wing mechanism to transition to the deployed flight configuration
- Integrated a torsion spring to generate wing rotation torque
- Incorporated nylon hardware and a locking mechanism to secure wing positioning



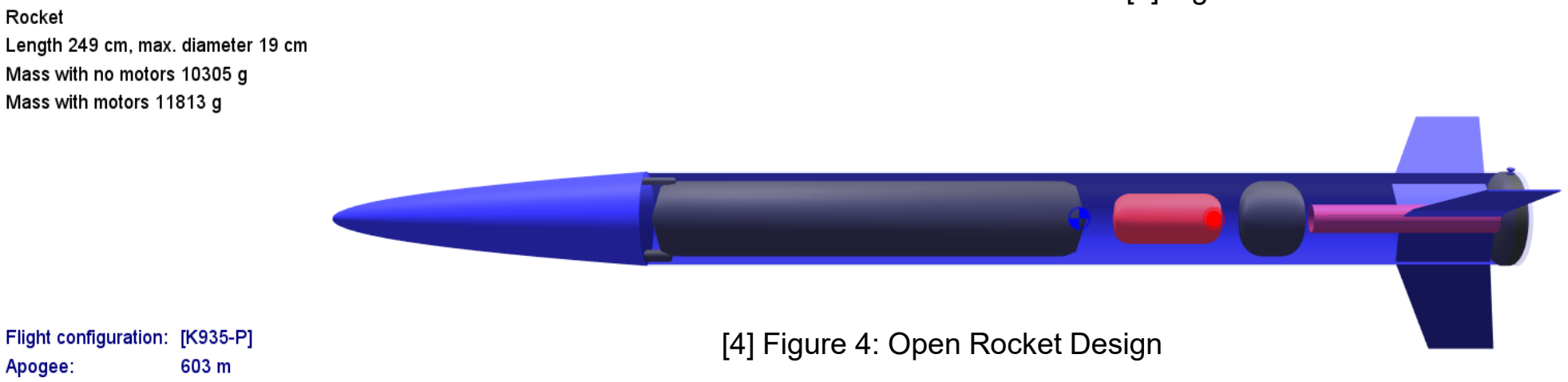

[2] Figure 2: UAV 3-View Flight Orientation Drawing

# Rocket Progress

- Rocket Design (Open Rocket)
- Rocket Body Part Ordering
- Initial On-Board Designs
  - Ejection Systems
  - Flight Computer / Telemetry
  - External Camera




[3] Figure 3: Altitude Versus Time Sim




[4] Figure 4: Open Rocket Design

# References

[1] Figure 1: UAV 3-View Launch Orientation Drawing made from OnShape.

[2] Figure 2: UAV 3-View Flight Orientation Drawing made from OnShape

[3] Altitude versus time graph of the rocket launch made with Open Rocket

[4] Figure 4: Open Rocket Design

# Thank You